Eli Doggart

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Experience

Nimbus Power Systems
Mechanical Design and Manufacturing Engineer **May 2022–Present**

- Designed, optimized, fabricated, and scaled up to prototype a four-fluid hydrogen fuel cell implementing performance, efficiency, and cost saving improvements (see projects for detail).
- Set up Haas vf 2 with custom vacuum chuck fixture to machine flat graphite sheets with tolerances down to +0.001-0" without the use of coolant, reduced cycle time from 45 minutes to 7.5 minutes
- Designed, built, wired, and plumbed 3 test stands for small-scale and engine-scale fuel cell testing, addressing safety for 1,000-volt wiring and flammable pressurized gases.
- Designed and built a facility-wide hydrogen safety system with emergency stops and sensors that trigger gas shutoffs, alarms, and facility exhaust fans.

Accelerated design LLC | *Design Engineering Intern* **January–March 2022**

- Revised CAD models, created drawings, operated SLA 3D printers.

General Dynamics Electric Boat | *Engineering Intern* **June–August 2021**

- Reviewed supplier requests to deviate from design for the pumps department for Columbia class.

Mott Corporation | *R&D Co-op* **January–June 2021**

- Ran testing equipment, improved lab procedures, conducted studies with metal DMLS printer tuning it for intentional porosity improving repeatability.

East Coast Precision Manufacturing | *Machine operator* **December 2020–January 2021**

- Operated various CNC lathes and mills to produce polymer parts.

Nomad Metallurgy | *Metallurgical Lab Assistant* **June 2016–August 2020**

- Metallurgical cut/mount/polish, microscopy, and hardness testing.

Niantic Bay Bicycles | *Bike Mechanic* **May 2016–August 2020**

- fixed and sold bikes, maintained stock, assisted with office tasks.

Selected Projects

Fuel cell design and fabrication | Nimbus Power Systems **2023-2026**
Designed, simulated, and optimized a 4 fluid hydrogen fuel cell for automotive and aerospace applications. Constrained by manufacturing limitations and legacy tooling, I reduced the reactant flow maldistribution from over 40% to less than 2%. Used flow and structural simulation to balance active area pressure and electrical resistance despite a reduction in landing area. After designing this fuel cell, I worked with component suppliers across the globe as well as scaled up in house procedures and facilities in order to produce an engine sized prototype.

Capstone Project | WPI **M3arch–December 2022**
Worked as designer and fabricator on team to produce, and optimize an active suspension system for a longboard.

Camper Van Build | Home **2025-Present**
Designed and fabricated a camper build for a Ram ProMaster including an off grid electrical system, plumbing, innovative bed and floor design. It is modular and easily adaptable due to use of extruded aluminum profiles.

Technical Skills

Design: SOLIDWORKS (CAD, Flow Simulation, CAM), Fusion (CAD CAM), LabVIEW, GD&T
Manufacturing: CNC and manual lathe and mill, 3D printing (FDM, SLA, DMLS), Basic MIG/TIG welding, wiring
Certifications: Six Sigma Online Black Belt, Trimech SOLIDWORKS Flow, Summit Toyotalift Forklift

Education

Worcester Polytechnic Institute (WPI), Worcester, MA **May 2023**
Bachelor of Science in Mechanical Engineering, GPA 3.63